Which Texas Bird Is It? Speaker Scripts

DATA 4380 · Mateen Jan, Tej Govindu, Alejandro Alfonso · Dr. Masoud Rostami · November 18, 2026

How to use this: For each slide: SAY is the script (also in the slide's speaker notes), WHY WE DID IT is the reasoning so you can explain it in your own words, and IF ASKED covers likely questions. Practice out loud with a timer: each person should land between 7 and 10 minutes (21 minutes minimum for the group).

Speaker	Part	Slides	Words	Time
Mateen	Problem and data	1–7	1,088	about 8 min
Tej	Baseline and improving the model	8–13	1,096	about 8 min
Alejandro	Results, interpretation and conclusion	14–21	1,100	about 8 min

Key numbers everyone should know: 1,710 photos · 10 species · split 1,197 / 256 / 257 · majority guess F1 0.02 · baseline CNN validation F1 0.61 (test 0.68) · augmentation 0.61 → 0.56 · MobileNetV2 0.956 (with aug.) / 0.964 (without) · fine-tuned 0.960 · final: frozen MobileNetV2, no augmentation: test accuracy 96.1%, macro F1 0.964 · 10 test mistakes, all in lookalike pairs (5 hawks, 4 crow/grackle, 1 dove) · baseline 81 mistakes · hawks ≈ 1/3 sky in borders · Grad-CAM border share 17–34% vs. 49% of area.

Slides 22–23 are backup slides (per-species F1, training curves) for questions.

Mateen: Problem and data (slides 1–7, about 8 min)

Slide 1 · Title

SAY

Good afternoon. We're Mateen, Tej and Alejandro, and our project is called Which Texas Bird Is It?

We trained a computer-vision model to look at one photo and name which of 10 Texas bird species it shows, including three pairs of birds that look almost the same.

I'll cover the problem, the data and how we prepared it. Tej will walk through our baseline and how we improved it, and Alejandro will cover the results, what the model actually looks at, and what we learned.

WHY WE DID IT

- Title slide items required by the guidelines: title, names, course, instructor, date.
- Saying the speaker plan up front helps the audience follow the hand-offs.

Slide 2 · Background and motivation

SAY

Identifying a bird from a photo is harder than it sounds. Anyone can tell a cardinal from a roadrunner, but two hawks, or a crow and a grackle, look almost the same at a glance. That's the problem we're solving: given one photo, name the species.

It matters because bird sightings are real data. Citizen-science apps like eBird and iNaturalist collect huge numbers of photos from regular people, and ecologists use them to track where species live and how populations change. A misidentified bird is a bad data point.

It helps beginner birders and students learning their local birds, the citizen-science projects that need clean records, and researchers sorting large photo collections.

We chose it because it's visual and fun, it uses the CNNs and transfer learning from this unit, and the lookalike pairs make it a real challenge instead of an easy yes-or-no.

WHY WE DID IT

- Plain-language problem first, before anything technical (guideline for slide 2).
- Covers all five guideline questions: what problem, why important, who benefits, real-world context, why this topic.

IF ASKED

- **Why only 10 species?** To keep the project doable on a laptop in the time we had, and to focus on Texas birds with three built-in lookalike pairs. Adding species is future work.

Slide 3 · Previous work

SAY

Four pieces of earlier work shaped our project.

Merlin Photo ID, from the Cornell Lab of Ornithology, is a deep neural network trained on millions of birders' photos. It also uses where and when the photo was taken, and it gets the right bird in its top three about 90% of the time. Lesson: even the best system needs help on hard species.

CUB-200-2011 is the standard research benchmark for telling similar birds apart, with 200 species. Its original baselines got only 10 to 17% accuracy, which shows how hard fine-grained bird ID is.

Projects on our exact dataset fine-tuned large ImageNet models and reported about 99% accuracy, because the photos are clean and centered.

And research on shortcut learning shows models can score well for the wrong reason, like a famous wolf-versus-husky classifier that was really detecting snow.

So compared with these, we use a small, fast pretrained model, we focus on the lookalike pairs, and we check what the model actually looks at.

WHY WE DID IT

- Two to four well-explained examples beat a long list (guideline).
- Each example connects to a decision we made: small pretrained model (BIRDS 525 projects), lookalike focus (CUB), Grad-CAM check (shortcut learning).

IF ASKED

- **Why is Merlin only ~90% top-3 if we got 96%?** Merlin handles hundreds of species and messy real-world photos; we have 10 species and clean, cropped photos. The numbers aren't comparable.
- **What is fine-grained classification?** Telling apart classes that look very similar, like species within the same bird family.

Slide 4 · Problem statement and objectives

SAY

Our question: can a convolutional neural network identify 10 common Texas bird species from a single photo, and does it struggle most with birds that look alike?

The target is the species. It's multi-class classification with 10 classes, one prediction per photo. Our main metric is macro F1, the F1 score averaged over all 10 species, so every bird counts equally.

We set four objectives: build a baseline before improving anything; improve one step at a time and report what each step does, including the ones that don't help; evaluate the final model once on held-out test photos; and check where the mistakes happen and what the model looks at.

Success means clearly beating the baseline, and being able to explain the mistakes the model still makes.

WHY WE DID IT

- The rubric checks that the question, target, task type and 2–4 objectives are all stated clearly.
- Macro F1 instead of accuracy: it weights each species equally and drops if any one bird is ignored.
- Objective 2 explicitly promises to report steps that don't help; the rubric rewards that.

IF ASKED

- **Why macro F1 and not accuracy?** Our classes are fairly balanced, so accuracy is meaningful too, but macro F1 drops if the model ignores any one species, which is what we care about.
- **What counts as success?** Clearly beating the baseline and being able to explain the remaining mistakes.

Slide 5 · Dataset and data understanding

SAY

Our data is BIRDS 525, a public image dataset of 525 species by Gerald Piosenka. The original Kaggle page has been taken down, so we used an identical public copy on Hugging Face. It's licensed CC0, which is public domain.

We kept 10 species that live in Texas: 1,710 photos. In table terms, each row is one photo. Its features are the pixels: 224 by 224 by 3 colors, about 150,000 numbers. The one label is the species. There's no missing data: every image opened and every image has a label.

The chart shows photos per species: 142 to 212 each, so the classes are fairly balanced, unlike our Dallas project.

On the right are our quality checks: no broken files and no exact duplicates. We also flagged 22 outliers: unusually dark, bright or high-contrast photos. We looked at every one. They're real photos against night skies, snow or bright sky, not errors, so we kept them.

That also matters for cleaning: in our Dallas project we had to drop over 124,000 rows with no due date. Here every photo is usable, so cleaning was about checking quality, not filling gaps.

WHY WE DID IT

- Guideline slide 5 asks for source and license, rows and columns, target, feature types, missing values, duplicates, class balance.
- For images: rows = photos, features = pixels, no categorical features, no missing values.
- Outliers: we inspected and kept them (the rubric asks how outliers were handled, not that they be removed).

IF ASKED

- **Why did you use a Hugging Face copy?** The original Kaggle page was removed. The Hugging Face mirror has the identical files, with the same CC0 license.
- **Why keep the outliers?** They're real photos in unusual light. Removing them would make the model worse at exactly those conditions.
- **Is there class imbalance?** Mild: 142 to 212 photos per species (ratio 1.49), so we didn't need resampling.

Slide 6 · Exploratory data analysis

SAY

Two things we found here shaped what we did next.

On the left are our three lookalike pairs: the Mourning Dove and Rock Dove, the Red-tailed and Red-shouldered Hawk, and the Crow and Common Grackle. A person can tell them apart by small details, like the grackle's yellow eye or the red-shouldered hawk's barred orange chest. We expected most of the model's mistakes to land inside these pairs.

On the right is what surprised us. We measured how much of each photo's border is blue sky. For both hawks it's about a third; for the Green Jay and Roadrunner it's 3 to 5%. A model could cheat by learning 'sky means hawk' instead of learning what a hawk looks like. That's called shortcut learning, and because of this chart we added heatmaps at the end to check where the model looks.

What we expected but didn't find: we expected a web-collected dataset to have duplicates or broken files. It had none.

We also checked whether the outliers favor one species. They're spread across all 10, with a few more among the dark birds, the crow and grackle, so they don't give the model an unfair hint about any one species.

WHY WE DID IT

- The guidelines want figures that changed what we did next: the sky chart directly caused the Grad-CAM check.
- Also covers 'what surprised you' (sky) and 'what you expected but didn't find' (duplicates/broken files).
- Sky share = fraction of the outer 16-pixel frame that is sky-blue (blue > red + 15, blue ≥ green, blue > 100).

IF ASKED

- **How did you measure 'sky'?** We counted pixels in the outer frame of each photo that are clearly blue (blue channel well above red). It's an approximation.
- **Why does background matter?** If one species is usually photographed against sky, a model can learn the sky instead of the bird. That's shortcut learning.

Slide 7 · Minimum preprocessing

SAY

Before the baseline, we did only what it couldn't run without.

First, the quality checks you just saw. Second, the split. The original dataset has only 5 test photos per species, so one mistake would swing accuracy by 2 points. We pooled all 1,710 photos and made our own stratified split: 70% train, 15% validation, 15% test, about 25 test photos per bird. Stratified means every species keeps the same share in each split, and a fixed random seed means anyone gets the same split.

Stratifying matters here because our smallest class, the Rock Dove, has only 142 photos. A purely random split could leave it with too few test photos to measure.

Third, for the baseline we shrank photos to 128 pixels so it trains quickly on a laptop, and scaled pixel values from 0–255 down to 0–1, which neural networks need.

Everything else we held back, so Tej can show what each step adds. And the rule throughout: training photos teach, validation photos choose, and test photos are used once, at the very end.

Hand off: "Tej?"

WHY WE DID IT

- Minimum preprocessing must be kept separate from later improvements, so each later step can be credited (guideline).
- Our own split: 5 test photos per class is too noisy; ~25 per class is more reliable.
- 128 px only for the from-scratch CNN (speed); MobileNetV2 later uses the full 224 px it was trained on.
- No leakage: the split was made once, by image, before any model; no image (by file hash) appears in two splits.

IF ASKED

- **Why make your own split?** The original test set had only 5 photos per species, so one mistake swung accuracy by 2 points.

- **Why 128 pixels for the baseline?** Speed on a laptop CPU. The pretrained model later uses the full 224 pixels it was trained on.
- **Any data leakage?** No. The split was made once by image before training, and we checked that no identical file appears in two splits.

Tej: Baseline and improving the model (slides 8–13, about 8 min)

Slide 8 · Baseline model

SAY

Thanks, Mateen. Our baseline has two parts.

First, the floor: always guess the most common species, the Red-tailed Hawk. That gets 12.5% accuracy and a macro F1 of 0.02. Any real model has to beat this.

Our real baseline is a simple CNN, a convolutional neural network, the standard starting point for images. It has four convolution blocks. Each one slides small filters over the photo to find edges, then textures, then shapes like beaks and wings, and pooling shrinks the image between blocks. At the end, a softmax layer gives a probability for each of the 10 birds.

A CNN assumes nearby pixels are related, and that a pattern means the same thing anywhere in the photo.

We use macro F1 because it averages all 10 species equally, so a model can't score well by ignoring one bird.

The baseline reaches a validation macro F1 of 0.61, about 61% accuracy. It's clearly learning, but it still gets about 4 in 10 birds wrong.

How we trained it: the Adam optimizer, up to 30 epochs, where one epoch is one pass through all 1,197 training photos, and early stopping that watches validation loss and keeps the best version. At the end, training accuracy was 68% and validation 63%. Those are close, so it isn't overfitting. If anything it was still learning when it hit the limit.

WHY WE DID IT

- The rubric weights the baseline heavily: it's what makes later numbers meaningful.
- Two baselines: majority guess (the floor) and a simple CNN (the simplest model that actually learns from pixels).
- Logistic regression on 49,152 raw pixels ignores image structure, so a small CNN is the defensible image baseline.
- The CNN was still improving at 30 epochs (train 68%, val 63%): somewhat under-trained, not overfit.

IF ASKED

- **Why not logistic regression as the baseline?** It would treat 49,152 pixels as unrelated columns and ignore the image's structure; a small CNN is the standard simplest image model.
- **Was the baseline overfitting?** No. Train and validation accuracy were close (68% vs. 63%); it was still improving at 30 epochs, so it was more under-trained.

Slide 9 · Additional preprocessing: augmentation

SAY

Our first improvement attempt was data augmentation: randomly flipping, rotating, zooming and changing the contrast of training photos, so the model sees more variety. Each choice fits birds: a bird facing left is the same species facing right, and birds are photographed at angles and in different light. It only applies to training photos.

Same CNN, same settings; augmentation was the only change. It didn't help: validation F1 went down, from 0.61 to 0.56.

To keep it fair, everything else stayed identical: same random seed, same architecture, same optimizer, same early stopping. On the training curves, in our backup slides, its validation accuracy bounced between about 48 and 55% while training accuracy kept climbing slowly.

Our explanation: with only about 120 photos per bird, a small network learning from scratch already struggles, and random changes make the task harder. It would need much longer training to benefit.

Later we ran the same check on the pretrained model: 0.956 with augmentation and 0.964 without. So augmentation didn't help either model, probably because these photos are already clean and centered. We didn't expect that, and we kept it in.

WHY WE DID IT

- Guideline slide 9: report what you did, why, and what happened to the metric, including steps that didn't help.
- Only one change between steps 1 and 2 (augmentation), so the drop can be credited to it.
- The MobileNetV2 check (with vs. without augmentation) is why the final model has no augmentation.

IF ASKED

- **Why do you think augmentation hurt?** The from-scratch CNN already struggled with ~120 photos per bird; random changes make learning harder and need many more epochs. For MobileNetV2 the photos are already clean and centered.
- **Did you apply augmentation to test photos?** No, only to training photos. Validation and test photos are always unchanged.

Slide 10 · Feature engineering: transfer learning

SAY

In a table, feature engineering means creating columns by hand. With images, a CNN learns its own features from the pixels. So our feature-engineering decision was where those features come from.

Our baseline learned its features from scratch, from about 1,200 photos. MobileNetV2 has already learned general visual features, like edges, textures, feathers and eyes, from about 1.3 million ImageNet photos, including many birds. Reusing them is called transfer learning.

We removed MobileNetV2's original 1,000-category output layer, froze everything else, and trained only a new 10-bird layer: 12,810 weights instead of the baseline's 242,000.

That one change took validation F1 from 0.61 to 0.96. Same photos, same split; the only difference is where the features come from. It's by far the biggest gain in the project.

Why does this work so well? The early layers of any image network learn general things, like edges, color blobs and textures, that are useful for almost any photo. Only the last step has to learn which combination of those means 'grackle' instead of 'crow'. And because those layers are frozen, they can't overfit to our small dataset.

Features we did not add: location and date, which Merlin uses. That's a future-work idea.

WHY WE DID IT

- Images have no hand-made columns; the meaningful 'feature' decision is learned-from-scratch vs. pretrained features.
- Frozen transfer learning trains only the new output layer (12,810 weights), which is why it works with ~120 photos per bird.
- The gain (0.61 → 0.96) is attributable because photos, split and metric stayed the same.

IF ASKED

- **What is ImageNet?** A dataset of about 1.3 million labeled photos in 1,000 categories that many vision models are pretrained on, including many bird categories.
- **Isn't that cheating?** No, it's standard practice. We still train the 10-bird output layer, and we compare against a from-scratch baseline so the gain is visible.

Slide 11 · Model selection

SAY

We compared two families of models. Why not logistic regression or a random forest? On raw pixels they treat each pixel as an unrelated column and ignore the image's 2-D structure, so CNNs are the standard choice for photos.

The simple CNN is the most transparent model, small enough to train from scratch, and it's our honest baseline. Its weakness is that it needs far more data than we have. It needs pixels scaled to 0–1.

MobileNetV2 is small and fast, built for phones, so it trains on a laptop in about 4 minutes, and it brings pretrained features. Its weakness: it assumes ImageNet features transfer to our photos, and it inherits ImageNet's biases. It needs full 224-pixel images scaled to minus 1 to 1.

We ran MobileNetV2 two ways: frozen, where we train only the new output layer, and fine-tuned, where we also retrain its top layers with a very small learning rate.

To keep the comparison fair, every model was trained the same way: the Adam optimizer, cross-entropy loss, and early stopping on validation loss. Only the model changed.

WHY WE DID IT

- Guideline slide 11: why each model, strengths, limitations, preprocessing needs. Not just a list of names.
- Assumptions: CNNs assume local, position-independent patterns; transfer learning assumes ImageNet features carry over.
- MobileNetV2 chosen over bigger models (ResNet, EfficientNet) because it trains in minutes on a laptop CPU.

IF ASKED

- **Why MobileNetV2 and not a bigger model?** It trains in minutes on a laptop CPU and was already at 96%. A bigger model might add a point, at much higher cost.
- **What are the CNN's assumptions?** Nearby pixels are related, patterns mean the same thing anywhere in the photo, and training and test photos come from the same kind of source.

Slide 12 · Model improvement

SAY

This chart shows every step on the validation set, and each bar changes one thing.

From left: guessing, 0.02. The baseline CNN in orange, 0.61. Adding augmentation, down to 0.56. Switching to pretrained MobileNetV2, up to 0.956. Removing augmentation, 0.964, the highest. And fine-tuning the top layers, 0.96, no better.

So we can credit the improvement to one decision: transfer learning. Two of our attempts didn't help, augmentation and fine-tuning, and we report both.

Fine-tuning not helping surprised us. Early stopping ended those runs after about 4 epochs, which means validation loss started getting worse right away. With only 1,200 photos, unfreezing 1.5 million weights gives the model room to memorize instead of learn.

You can see the opposite with the frozen model in our backup slide: its training and validation accuracy stay right together, around 95%, which means it isn't overfitting at all.

WHY WE DID IT

- 'A rising line is only convincing when each step is attributed to a specific decision' (guideline slide 12).
- Every bar is on the same validation set with the same metric.
- Early stopping after ~4 epochs = validation loss rose after epoch 1 (patience 3 → best epoch was the first).

IF ASKED

- **Why didn't fine-tuning help?** With ~1,200 photos, unfreezing 1.5 million weights lets the model memorize; validation loss rose after the first epoch.
- **Is 0.956 vs. 0.964 a real difference?** It's about 2 photos out of 256, so we treat it as 'augmentation gave no benefit', not 'augmentation is harmful'.

Slide 13 · Hyperparameter tuning

SAY

We tuned the fine-tuning step, because that's where the settings matter most. Two hyperparameters: how many of MobileNetV2's top layers to unfreeze, 30 or 60, because more layers means more flexibility but more risk of overfitting; and the learning rate, 1e-5 or 3e-5, because too high a rate wrecks the pretrained weights.

We used a grid search, all four combinations. Each started from the same frozen model and was scored on the validation set only. The test set was never used to choose anything.

Why grid search and not randomized search? With only four combinations, we can simply try them all. Randomized search makes sense when there are too many combinations to try.

Before and after: the default, 30 layers at 1e-5, scored 0.960, and it was also the best combination, so tuning changed nothing. The higher learning rate hurt: 0.93 to 0.94.

Tuning doesn't guarantee improvement, and here it confirmed that the simpler frozen model was already enough.

Hand off: "Alejandro?"

WHY WE DID IT

- Tuning must use validation data only; show before/after (guideline slide 13).
- Grid search is fine for 4 combinations; randomized search makes sense only for bigger spaces.

- Each run starts from the same frozen weights, so the comparison is fair.

IF ASKED

- **Why only four combinations?** Each run takes a few minutes on a laptop; four covered the two settings that matter most for fine-tuning.
- **Did you tune the frozen model?** Not separately; it already scored 0.964 with standard settings, and the fine-tuning grid shows extra flexibility didn't help.

Alejandro: Results, interpretation and conclusion (slides 14–21, about 8 min)

Slide 14 · Model evaluation and comparison

SAY

Thanks, Tej. Here's every model on the same 257 test photos, which none of them trained on.

We chose the final model before looking at this table, using validation macro F1, with one rule set in advance: if models are within 0.005 of each other, less than one photo, keep the simpler one. MobileNetV2, frozen, without augmentation, had the highest validation F1 and the fewest trained weights, so it's our final model.

A quick guide to the columns: accuracy is the share of photos labeled correctly. Precision asks: when the model says 'crow', how often is it right? Recall asks: of the real crows, how many did it find? F1 balances the two, and ROC-AUC measures how well it ranks the right species above the others. All of them are averaged over the 10 species.

On test it gets 96.1% accuracy and a macro F1 of 0.964, compared with 0.68 for the baseline CNN and 0.02 for guessing. Precision and recall are both about 96%, so it isn't trading one for the other.

One row worth noticing: MobileNetV2 with augmentation scored 0.956 on validation but 0.936 on test. With only about 25 photos per species, scores can move a couple of points between sets, so we don't read too much into small gaps.

Test scores line up with validation for every model, so our choices weren't overfit to the validation set. The fine-tuned model also does well here, 0.955, but switching now would mean choosing by the test set.

WHY WE DID IT

- Compare every model on the same held-out data, report the poor ones too, and justify the final pick (guideline slide 14).
- Selection rule set before looking at test: highest validation macro F1, ties (≤ 0.005) go to the simpler model.
- ROC-AUC is one-vs-rest, averaged over the 10 species.

IF ASKED

- **Why not pick the fine-tuned model since it's close?** It scored lower on validation, which is what we choose with. Switching after seeing test scores would be choosing by the test set.
- **Is 96% believable?** On these clean, centered photos, yes, and it matches published work on this dataset. It would be lower on real backyard photos.

Slide 15 · Interpretation: what it gets wrong

SAY

This is the final model's confusion matrix on the test set: the true species down the side, the predicted species across the top. Almost everything is on the diagonal.

It made 10 mistakes out of 257, and every one is inside a lookalike pair: 5 between the two hawks, 4 between the crow and the grackle, and 1 between the doves. No random mistakes, like calling a cardinal a hawk. The baseline made 81 mistakes, and 49 of those were outside the pairs.

Per species, four birds are perfect on the test set: the cardinal, green jay, roadrunner and mockingbird. The lowest are the two hawks and the black birds, around 0.91 to 0.93, exactly the pairs we predicted back on slide 6.

On the right are some of them, and they're understandable: a hawk seen from behind hides the chest and tail colors that separate the species, and a grackle in dull light loses its purple sheen and looks like a crow.

7 of the 10 mistakes had confidence under 65%, so the model is usually unsure when it's wrong. A real app could say 'not sure' instead of guessing.

WHY WE DID IT

- Guideline slide 15: what the model gets wrong, with a confusion matrix and error analysis.
- Answers objective 4 directly: errors fall inside the pairs we predicted in EDA (slide 6).
- Low confidence on errors suggests a practical 'not sure' threshold.

IF ASKED

- **Which pair is hardest?** The two hawks: 5 of the 10 mistakes.
- **What would you do about low-confidence predictions?** Show 'not sure' or the top 2–3 guesses below a confidence threshold, like Merlin does.

Slide 16 · Interpretation: where it looks (Grad-CAM)

SAY

Remember the sky problem: hawks are photographed against sky far more often. So we checked what the model actually looks at, using Grad-CAM. It colors the parts of the photo that pushed the model toward its answer; red means most important.

The top row is the original photo, and the bottom row is the same photo with the heatmap on top.

For the hawks, crow and grackle here, the heat sits on the bird's body: the chest, the tail, the head. That's the plumage that really separates these species.

We also measured it across all 257 test photos. The outer frame is about half of each image, but it gets only 17 to 34% of the heatmap for every species, including both hawks. So the model mostly looks at the bird, not the background.

One exception, on the far right: for this rock dove it focused on the corner. That bird also looks like a different pigeon species, which points to some label errors in the data.

WHY WE DID IT

- Closes the loop on the EDA finding (sky backgrounds) with evidence.
- Border share: outer 32-pixel frame = 49% of the image area; heatmap weight there is 17–34% for every species.
- Grad-CAM shows where, not why; importance is not causation (guideline note).

IF ASKED

- **How does Grad-CAM work?** It looks at how much each region of the last convolution layer pushed up the score of the predicted species, and turns that into a heatmap.

- **Does this prove the model isn't using background?** No, it reduces the concern. Heat in the center doesn't prove why the model decided, only where the evidence came from.

Slide 17 · Key findings

SAY

Back to our four objectives.

One, the baseline: a simple CNN reached a macro F1 of 0.61, far above guessing, but it got 4 in 10 birds wrong.

Two, what helped: transfer learning gave essentially the whole improvement, 0.61 to 0.96. Augmentation and fine-tuning didn't help.

Three, the final model, frozen MobileNetV2, names 96% of test photos correctly, with a macro F1 of 0.964.

Four, its mistakes are all between lookalike species, the same pairs a beginner birder would mix up, and it looks at the bird rather than the sky.

So yes, a CNN can identify these 10 Texas birds from one photo, and when it fails, it fails for understandable reasons. The insight: with small data, reusing what a big model already learned beats anything we tuned ourselves.

WHY WE DID IT

- Key findings must answer the slide-4 objectives, as insight rather than a second metrics table.
- One sentence per objective, plus the overall answer to the research question.

IF ASKED

- **What was the most important decision?** Using pretrained MobileNetV2 features: 0.61 to 0.96 on validation.

Slide 18 · Limitations

SAY

We want to be clear about what this does not show.

First, the photos are clean, cropped and centered, and many come from the web. Real backyard or trail-camera photos are blurrier and farther away, so 96% here does not mean 96% in the field.

Second, the model only knows 10 species, and it always picks one of them. Show it a penguin and it will confidently name a Texas bird.

Third, label noise: a few photos look mislabeled, like a 'mourning dove' with a collared dove's neck ring.

Fourth, a small test set and one split: about 25 photos per species, so one photo moves a species' score by about 4 points.

And finally, we inherit ImageNet's biases, and Grad-CAM shows where the model looks, not why.

How we'd address these: test on real Texas photos, re-check suspicious labels starting with the photos the model gets confidently wrong, and repeat the split a few times to report a range instead of one number.

WHY WE DID IT

- Honest, specific limitations: dataset size, sampling bias (web photos), label noise, generalizability, single split.

- 'The conclusion states what the result does and does not support' (rubric).

IF ASKED

- **How would you fix label noise?** Re-check suspicious labels by hand, starting with the photos the model gets confidently wrong.
- **Would it work in another state?** Only for these 10 species; birds elsewhere would need new training photos.

Slide 19 · Conclusion, future work and what we learned

SAY

To sum up: we asked whether a CNN can tell 10 Texas birds apart from one photo. Our final model, a frozen pretrained MobileNetV2, gets 96% of test photos right, up from 68% for our baseline, and its only mistakes are between birds that genuinely look alike.

In practice, a model like this could help beginners and citizen-science apps, as long as it says 'not sure' when its confidence is low.

For example, below 65% confidence it could show its top two or three guesses instead of one answer. That alone would have covered 7 of our 10 mistakes.

Next steps: test it on real Texas backyard photos, add more species and an 'unknown bird' option, and use location and date the way Merlin does.

What we learned: a bigger or more complicated model isn't automatically better; the simplest transfer-learning model won. Steps we were sure would help, like augmentation, didn't. And a high score isn't enough: you have to check what the model is looking at.

WHY WE DID IT

- Conclusion: problem, most important finding, final model, practical meaning, next steps (guideline slide 18).
- 'What we learned' answers the course reflection: how the project challenged our assumptions.

IF ASKED

- **What would you do with more time?** Collect real Texas backyard photos to test it, and add an 'unknown' option.

Slide 20 · References

SAY

Our references are here in APA style: the dataset, the MobileNetV2 and Grad-CAM papers, the benchmarks we compared against, and our code on GitHub.

WHY WE DID IT

- One citation style (APA) for everything: dataset with version and license, papers, tools.

Slide 21 · Questions

SAY

Thank you. We're happy to take any questions.

WHY WE DID IT

- Simple closing slide; backup slides sit behind it for questions.